

Design Verification Plan and Report (virtual test) — VBF-T2R1FLASH-DVPR-01

****Case**:** t2_r1_flash · **Verdict:** ****ALL PASS**** (6/6 criteria)

#	Verification item	Condition	Result	Determination (vs criteria)	Source
1	1C discharge capacity	1C CC discharge 25C to 2.5 V, DFN	5.0469 Ah / 18.178 Wh	PASS (ED >= 327.18: 425.4 Wh/kg)	cell/r2_fcc_1c_dfn.json, r2_fcc_energy.json
2	4C fast-charge temperature rise	4C CC charge 45C amb, lumped thermal	T_max 329.2 K (red line 333.15 K)	PASS	cell/r2_fcc_4c.json
3	4C fast-charge plating	4C charge + plating module	anode min 12.3 mV >= 0 V	PASS (no plating)	cell/r2_fcc_4c.json anode_potential_v
4	SEI growth @100 cyc 1C	1C CC cycling 100 cyc, SPMe aging	78.2 nm	PASS (<= 500 nm)	cell/r2_fcc_aging100.json
5	SEI growth @500 cyc 1C	1C CC cycling 500 cyc, SPMe aging	262.0 nm	PASS (<= 550 nm)	cell/r2_fcc_derived.json
6	Low-temperature discharge	1C discharge -20C	99.4 % of 25C capacity	PASS (>= 90 %)	cell/r2_fcc_derived.json
7	Voltage window	parameter set	2.5 – 4.2 V	PASS (contract window)	parameter set

****Items marked N/A (beyond pure simulation boundary, requires physical experiment)**:** nail penetration, overcharge-to-thermal-runaway, crush, drop, rate-pulse internal resistance, cycle life to 80% SOH (capacity-based), cell-can/tab mechanicals, formation optimization, manufacturing tolerances.

****Known artifact (honest note)**:** under the standard SEI model the aging capacity trajectory shows a climb-then-saturate non-monotonic shape (lithium-loss shifts the voltage window); it is NOT treated as normal degradation — the reliable aging indicator is sei_thickness_nm_end (used for verdicts 4/5).